

TUNNAK, Australia

WMO: 949600

Lat: 42.454S

Lon: 147.461E

Elev: 1516

StdP: 13.91

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.2	30.3	26.0	21.2	32.4	27.5	22.7	34.0	32.7	46.4	28.7	45.6	2.7	120	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.1	80.1	61.1	75.9	60.1	72.3	58.7	63.8	74.0	62.1	71.5	60.5	69.1	16.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	83.4	65.8	58.7	77.8	64.9	56.9	73.0	63.5	29.9	74.1	28.6	71.6	27.4	69.4	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.4	24.8	22.0	DB	24.6	89.4	1.8	5.0	23.3	93.0	22.3	95.9	21.3	98.7	20.1	102.4	(4)
(5)			WB	24.2	67.1	1.7	1.5	23.0	68.2	22.0	69.0	21.0	69.9	19.7	71.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	58.6	58.1	55.4	50.0	46.0	42.7	41.8	42.7	45.5	48.4	52.2	55.3
	DBStd	7.85	6.20	5.98	6.17	5.31	4.75	4.12	3.59	4.13	4.41	5.37	5.83	5.82
	HDD50	1235	4	5	18	64	142	222	254	229	148	96	39	14
	HDD65	5641	216	205	306	450	590	670	719	692	586	516	385	306
	CDD50	1117	272	232	184	64	18	2	0	2	11	46	106	180
	CDD65	45	18	12	7	0	0	0	0	0	0	0	2	6
	CDH74	568	247	125	72	1	0	0	0	0	0	5	38	80
	CDH80	136	72	24	19	0	0	0	0	0	0	0	4	17
Wind	WSAvg	10.4	10.8	9.9	10.2	9.7	9.9	9.2	10.0	10.6	11.7	11.3	10.7	11.1
Precipitation	PrecAvg	22.5	1.9	1.7	1.5	1.7	1.6	1.9	1.9	2.2	2.1	1.9	2.1	2.1
	PrecMax	31.1	4.6	4.4	3.4	4.6	4.9	5.9	3.6	5.0	5.3	3.7	5.2	6.2
	PrecMin	14.4	0.3	0.3	0.4	0.5	0.3	0.4	0.7	0.4	0.8	0.3	0.2	0.2
	PrecStd	4.3	1.1	0.9	0.9	0.9	0.9	1.4	0.7	1.1	1.0	0.7	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.5	83.8	82.3	70.6	61.7	56.1	53.9	56.8	62.8	73.3	79.0	82.7
		MCWB	63.5	62.6	61.3	57.0	53.2	50.8	48.6	49.5	52.9	55.5	59.6	62.9
	2%	DB	80.6	77.9	74.3	65.7	58.3	53.5	51.7	54.1	58.6	65.4	72.0	75.4
		MCWB	61.5	61.3	60.0	55.3	51.7	48.9	46.9	47.8	50.8	54.2	57.8	60.0
	5%	DB	75.3	73.6	69.9	62.3	55.8	51.4	49.9	52.0	55.9	61.8	67.3	71.0
		MCWB	59.6	60.0	58.1	53.9	50.5	47.5	45.8	47.0	49.1	52.6	55.9	57.8
	10%	DB	71.0	69.8	66.1	59.4	53.8	49.8	48.3	50.1	53.7	58.4	63.0	67.0
		MCWB	58.1	58.7	56.5	52.5	49.5	46.5	44.7	45.9	47.8	50.8	54.2	56.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.9	65.7	64.2	59.4	55.0	53.3	50.3	51.6	55.1	58.5	62.4	64.5
		MCDB	80.3	75.5	74.7	66.2	58.2	54.7	52.2	54.3	59.3	66.0	73.2	74.1
	2%	WB	63.6	63.8	61.8	57.2	53.1	49.9	47.8	49.4	52.3	55.9	59.4	61.7
		MCDB	75.1	72.7	69.9	62.6	56.4	52.0	50.3	52.4	56.6	62.8	68.1	71.5
	5%	WB	61.5	61.9	59.9	55.3	51.5	48.3	46.5	47.6	50.2	53.7	57.3	59.4
		MCDB	71.5	70.4	67.3	60.3	54.6	50.8	48.9	51.1	54.5	60.3	65.6	68.6
	10%	WB	59.5	60.0	58.1	53.5	50.1	46.7	45.2	46.1	48.4	51.5	55.0	57.3
		MCDB	68.9	68.0	64.7	58.3	53.2	49.3	47.8	49.5	52.9	57.5	62.0	65.6
Mean Daily Temperature Range	5% DB	MDBR	21.1	19.9	18.6	16.8	13.3	12.1	12.1	13.4	15.0	17.1	18.2	19.2
		MCDBR	29.1	26.0	23.5	20.6	15.3	13.4	13.5	15.2	17.5	21.7	25.0	25.7
		MCWBR	12.7	12.0	10.9	11.4	10.0	9.7	9.9	10.2	11.4	12.5	13.0	12.3
	5% WB	MCDBR	24.9	22.0	20.3	17.5	13.4	11.5	11.9	13.5	15.9	19.7	22.0	22.5
		MCWBR	12.6	11.9	10.3	10.6	9.3	9.1	9.3	9.8	11.6	12.2	12.3	12.1
Clear-Sky Solar Irradiance	taub	0.325	0.315	0.312	0.297	0.286	0.275	0.276	0.289	0.307	0.314	0.317	0.320	
	taud	2.520	2.571	2.568	2.592	2.601	2.616	2.617	2.584	2.518	2.508	2.524	2.527	
	Ebn at Noon	318	312	299	282	263	256	265	281	296	309	318	321	
	Edh at Noon	35	31	28	23	20	17	19	23	29	33	34	35	
All-Sky Solar Radiation	RadAvg	2134	1816	1379	938	588	463	532	789	1183	1618	1906	2128	
	RadStd	173	81	72	56	33	35	22	50	61	94	148	128	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) -1.24	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (33 neighbors)	+0.58	N/A	N/A	+1.19	N/A	N/A	N/A	-171	+185	+22

Nomenclature: See separate page